Hasin Ahmad Zafir

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SUMMARY

Final-year Mechanical Engineering student at UNSW specialising in engineering design, fluid dynamics, and simulation. Proven experience delivering technical outcomes across aerodynamics, manufacturing, and commercial engineering consultancies. Strong collaborator skilled in multidisciplinary team environments and digital workflows, seeking to contribute to mechanical and client-focused projects and develop toward professional accreditation.

EDUCATION

University of New South Wales, Mechanical Engineering *Feb 2027 (Expected)*
Relevant Courses: Refrigeration & HVAC, Noise & Acoustics, Finite Element Analysis, Computational Fluid Dynamics, Mechanics of Fracture and Fatigue.

PROJECTS

Aerostructures Engineer – UNSW Design Build Fly (DBF)

- Applied finite element analysis and composite material principles to design lightweight, high-efficiency aircraft structures
- Improved structural efficiency under strict competition constraints by reworking cargo placement and weight distribution
- Collaborated within a multidisciplinary engineering team, contributing to iterative design reviews and manufacturing phases

Design & Fabrication of Stirling Engine (UNSW)

- Designed and built a fully functional Stirling engine with a 400 rpm 300 gram flywheel, applying thermodynamics and mechanical system design principles
- Developed detailed CAD assemblies and multi-level BOM in SolidWorks, ensuring manufacturability and system integration across 24 sub-assemblies
- Delivered a working prototype within 10 weeks by conducting iterative testing and refinement, supported by technical reporting and design justification for STEM outreach and demonstration applications

Founder & Team Lead – Formula Student Team (Formula IUT)

- Founded a student engineering team, leading the end-to-end delivery of a Formula Student vehicle from concept to fabrication
- Designed and built a space frame chassis, with a factor of safety of 1.8 and a weight of 98 kg
- Coordinated subteams, including aerodynamics, suspension, and drivetrain, while managing a budget of $8000
- Achieved 3rd place at Formula Bharat 2023 (Class 1) against 53 competing teams

KEY EXPERIENCE

CPP Wind Engineering Consultants, Lab Engineering Intern | *Dec 2025 – May 2026*

- Designed 3 R&D projects using CAD and DFM principles to optimise lab testing environments and commercial models
- Fabricated and analysed wind-tunnel-ready aeroelastic models for commercial testing under variable deadlines and testing conditions
- Performed frequency analysis, damping analysis, and torque analysis on scaled solar panel arrays to evaluate model validity under wind loads
- Troubleshot test model setup and sensor connections to ensure high data accuracy during engineering analysis

ANTS Aerial Systems, Research and Development Executive | *Oct 2024 – Feb 2025*

- Utilised QGIS, PIX4D, WebODM, and other mapping tools to create digital 2D and 3D maps, and authored a process SOP for future onboarding
- Coordinated time and resources to deliver 3 detailed area maps covering 620 acres within 1 week, helping secure 3 government contracts valued over $25,000
- Developed and delivered operational tutorials for 2 corporate clients purchasing automated aerial solutions

Runner Motors Limited, Senior Executive (Inventory and Procurement) | *July 2024 – Oct 2024*

- Managed sorting, receiving, and delivery operations for a 12,000-product warehouse to improve supply chain reliability
- Engineered an adjustable hand trolley that reduced individual fabrication costs from roughly $180 to $56, representing a 68% cost reduction
- Applied analytical problem-solving to improve stock availability, reducing monthly lost sales by $7,200 per month, and processed $33,000 in warranty parts.
- Led daily collaborations within a 15-member team to improve floor efficiency, presenting outcomes via a new warehouse SOP

ADDITIONAL EXPERIENCE & VOLUNTEERING

UNSW Mechanical and Manufacturing School, Casual Academic (CFD) | *Feb 2026 – Present*

- Running technical laboratory sessions and interactive workshops, guiding 33 Bachelor's and Master's engineering students through ANSYS CFD workflows

Red Cross Shop, Volunteer member | *Feb 2026 – June 2026*

- Handle point-of-sale transactions to support store operations
- Navigating customer conversations and verbal feedback to ensure proper service satisfaction and repeat store visits

Texties Fashion Sourcing Co., Business Development Executive | *June 2024 – Jan 2025*

- Generated 2500 qualified leads via LinkedIn Sales Navigator and other sources, corresponding with outreach KPIs to secure 2 new corporate clients

PuzzleFun World, Product Development Intern | *Feb 2021 – May 2021*

- Designed, prototyped using laser cutting, and built a 3D puzzle of an architectural building as a toy to be sold by the company, along with its written guide and rendered photorealistic images to be used for the product catalogue.
- Conceptualised a rudimentary inventory management system through MS Excel to account for raw materials, material placements, and material usage.
- Demonstrated the ability to lead a team of 3 machine operators to meet the demand of 150+ product orders/month under challenging operational conditions and variable client demands

KEY SKILLS

Technical:

- <u>CAD & Simulation</u>: Solidworks (**Certified Solidworks Expert (Credential ID: C-MNXVL8ADY2)**), Rhinoceros, Autodesk Fusion, Ansys (FEA, CFD, and Composites), Keyshot, 3DExperience online, OpenFOAM
- <u>Mapping & Data Analysis</u>: QGIS, PIX4D, WebODM
- <u>Productivity</u>: MS Office Suite, Google Suite, SAP, Oracle NetSuite

Core Competencies:

- Engineering Problem Solving & Data Analysis
- Multidisciplinary Team Collaboration
- Communication & Stakeholder Engagement
- Process Optimisation & Continuous Improvement

INTERESTS

Learning new things, working out, travelling and hiking, meeting new people and cultures, and playing story-driven video games.